



Realistic human-like avatar embodiment diminishes outcomes in digital emotion regulation interventions

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ABSTRACT

Digital mental health prevention and therapy programs are increasingly integrating embodied conversational agents and avatars to improve user interaction and psychological well-being. However, the effects of human-like avatar embodiment remain underexplored, particularly in digital interventions targeting emotion regulation, a critical transdiagnostic factor of various mental health conditions. Thus, the current study investigates the impact of avatar realism in a brief online emotion regulation intervention.

In a between-subjects online experiment, 211 participants completed a single-session digital emotion regulation intervention, featuring psychoeducational content and practical exercises, facilitated by an embodied conversational avatar. Participants were randomly assigned to one of four conditions: realistic humanoid avatar, abstract humanoid avatar, robot avatar, or a control condition using audio waveform animations. Emotional ratings, satisfaction with the intervention, avatar user experience, self-disclosure levels, and open-ended feedback were assessed through self-report.

Positive emotions increased across all conditions, except for the realistic avatar group during a gratitude exercise. Results further revealed a significant main effect of condition on intervention satisfaction, with the realistic humanoid avatar receiving lower ratings than the other conditions. Interactions with the realistic avatar were rated as significantly more anxiety-inducing and less pleasant. Additionally, participants in the realistic humanoid avatar condition reported significantly lower levels of self-disclosure compared to the other conditions.

These findings suggest that realistic humanoid avatars may induce discomfort in emotion regulation interventions, potentially reducing their effectiveness. The results are discussed considering the uncanny valley and perceptual mismatch hypotheses, highlighting the need for careful multimodal avatar design in digital mental health interventions.

1. Introduction

The global mental health crisis has reached alarming levels, with approximately half of the population estimated to experience a mental health issue at some point in their lives (McGrath et al., 2023), highlighting the need for effective prevention and treatment strategies. As traditional mental health services struggle to keep up with growing demand, digital mental health programs, including semi- or fully automated apps, have emerged as promising supplementary solutions, offering scalable, low-barrier interventions to help address these challenges (Philippe et al., 2022).

A critical influential factor of many mental disorders is emotion regulation difficulties, which have been identified as a transdiagnostic process that underlies a range of psychiatric conditions across all ages

(Aldao et al., 2016; Schäfer et al., 2017). Emotion regulation encompasses the conscious and automatic processes that determine the timing, intensity, and expression of emotional responses (Gross, 2015). Although the effectiveness of specific regulatory strategies seems to partially depend on contextual factors (Aldao & Tull, 2015), research indicates that certain strategies are generally more adaptive, typically associated with well-being and reductions in negative affect, including accepting emotions as they arise, cognitively reappraising situations, or cultivating gratitude (e.g., Du & Liu, 2025). In contrast, other strategies such as emotional suppression or experiential avoidance are often considered maladaptive, as they tend to maintain or exacerbate emotional distress (Feldner et al., 2003). A substantial body of literature demonstrates that individuals with mental health conditions, including depression, anxiety disorders, substance abuse, or eating disorders,

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often exhibit dysfunctional emotion regulation profiles (e.g., Aldao et al., 2010; Hu et al., 2014), marked by reduced use of adaptive strategies and an increased reliance on maladaptive ones. Beyond correlational evidence, longitudinal and experimental studies suggest that maladaptive regulation strategies contribute to the onset and persistence of psychopathology (Chervonsky & Hunt, 2019; Dryman & Heimberg, 2018; Naumann et al., 2015), whereas greater use of adaptive strategies can buffer against symptom emergence and deterioration. Given this central role of emotion regulation, digital mental health prevention and intervention programs increasingly aim to strengthen adaptive strategies (Eisenstadt et al., 2021). By fostering skills such as cognitive reappraisal or gratitude practices, these programs seek to equip individuals with effective tools for managing emotional experiences and thereby reduce their vulnerability to severe mental health problems (Berking et al., 2022).

Despite the growing adoption of digital mental health services, a substantial gap remains in rigorous research evaluating their effectiveness (Bucci et al., 2019). Experimental investigations are specifically needed to identify optimal design elements for these interventions (Graham et al., 2019), as many programs encounter difficulties with user engagement, resulting in low adherence and high attrition rates (Fleming et al., 2018; Nwosu et al., 2022). Elevating user experience is crucial, given that research on digital mental health programs suggests higher engagement is linked to greater post-intervention benefits (Gan et al., 2021). Thus, developers and researchers are increasingly incorporating and exploring engagement-enhancing techniques and persuasive technology, such as conversational agents and chatbots—interactive features that simulate human-like communication to foster rapport and sustain user interaction (Gaffney et al., 2019; Sebastian & Richards, 2017; Sukanuma et al., 2018). Within this context, agents with graphical embodiments, such as animated avatar characters, are often designed and implemented with the intention of enhancing the perception of social presence. Due to their ability to visually mimic human behaviors more directly than text- or voice-only interfaces, they hold particular potential to foster a more human-like interaction experience, which may in turn support the development of a stronger user-machine relationship and contribute to positive mental health outcomes (Provoost et al., 2017). For example, applications like Woebot leverage a robot avatar to interact with users about their emotional well-being, illustrating how such features can contribute to effective digital mental health experiences (Durden et al., 2023).

In the framework of avatar embodiment, recent technological advancements have enabled the development of highly realistic digital humans, allowing for the display of lifelike facial expressions and offering the capacity for even more natural and engaging interactions in digital programs. Despite this, the impact of avatar realism on user experience and intervention effectiveness remains underexplored (Ter Stal et al., 2020), especially in the field of healthy emotion regulation and mental health. Research suggests that humans possess an innate tendency to engage more attentively with faces, demonstrating a distinct preference for human faces over those of monkeys as early as three months of age (Di Giorgio et al., 2013). On the other hand, the “uncanny valley” phenomenon suggests that robots and avatars approaching, but not fully achieving, human-likeness can provoke discomfort and eeriness (Mori et al., 2012).

Given the mixed empirical evidence on the uncanny valley, with some studies demonstrating its presence (e.g., Mathur et al., 2020; Song & Shin, 2024) and others reporting positive feelings to increased anthropomorphic embodiment (e.g., Liu & Wang, 2025; Ma & Li, 2024), a growing body of research has begun to identifying the specific features that reliably elicit negative responses towards human-like avatar embodiment (Di Natale et al., 2023; Kätsyri et al., 2015). Subtle inconsistencies in realism such as deviations in facial morphology or disproportionate features (e.g., overly large eyes) have repeatedly been shown to heighten feelings of eeriness (Burleigh et al., 2013; Chattopadhyay & MacDorman, 2016; Schwind et al., 2018). In addition,

dynamic anomalies in behavior, including unnatural facial motion, temporal delays, or reduced variability in micro-expressions seem to further intensify uncanny impressions (Cihodaru-Ştefanache & Podina, 2025). Notably, overly rigid upper-face dynamics lacking emotional expressivity (Tinwell et al., 2013; Tinwell, Grimshaw, Abdel-Nabi, et al., 2011) as well as insufficient emotional responsiveness (Rosenthal-von der Pütten et al., 2018) have been identified as contributors. Conversely, exaggerated or excessive emotional displays may also evoke discomfort (Tinwell, Grimshaw, & Abdel-Nabi, 2011). Beyond facial cues, vocal characteristics play a similarly critical role (Suma et al., 2023). Virtual human-like agents are perceived as more uncanny when their speech is asynchronous with facial movements, such as in cases of inaccurate lip-syncing (Tinwell et al., 2015). Thus, to mitigate these negative responses, research highlights the importance of equipping digital humans with naturalistic appearance and motion, combined with socially supportive behaviors (Cihodaru-Ştefanache & Podina, 2025; Zhang et al., 2020).

These findings are well explained by the Perceptual Mismatch Hypothesis, which suggests that human-like avatars raise strong expectations for aligned appearance and behavior, and that even subtle inconsistencies in realism, motion, expression, or responsiveness can violate these expectations and trigger perceptual discomfort (Kätsyri et al., 2015). Empirical work shows that ambiguous stimuli, those intermediate between non-human and human categories, elicit prolonged classification times and heightened eeriness, a pattern also observed in artificial neural networks (de Borst & de Gelder, 2015; Igaue & Hayashi, 2023; Sasaki et al., 2024; Wang & Rochat, 2017). Together, these scientific contributions suggest that expectancy violation, perceptual mismatch, and categorization uncertainty may jointly shape the uncanny-valley experience. Crucially, however, within these frameworks, highly anthropomorphic avatars with closely aligned appearance and movement, made possible by cutting-edge digital human technologies, may circumvent this discomfort, eliciting positive responses and supporting more natural human-avatar interactions, although empirical evidence remains limited (Kätsyri et al., 2015).

Beyond this, the context-specific effects of human avatar realism further complicate the picture. For instance, realistic humanoid avatars have been found to foster stronger user-avatar relationships and increase the intention to engage in the metaverse (Kim et al., 2023). Empirical evidence further indicates that they are also perceived as more suitable for medical tasks, whereas stylized, cartoon-like avatars seem to be favored in casual social interactions (Ring et al., 2014). Similarly, in educational settings, avatars with highly human-like embodiments have been shown to advance greater trust and a sense of closeness, leading to more positive learning experiences compared to less human-like, abstract alternatives (Chae et al., 2024). These findings highlight the duality of human avatar realism and raise questions about the optimal design of visually embodied agents for digital mental health interventions.

Building on this background, the present study experimentally explored the impact of avatar realism and humanoid embodiment on the outcomes of a brief digital emotion regulation intervention session for individuals seeking to improve their well-being and emotion regulation skills. More precisely, the study compared the effects of a realistic humanoid avatar as a virtual coach with three alternative conditions: an abstract toon-styled humanoid avatar, a robot avatar, and an audio waveform animation. This design enabled a systematic manipulation of human resemblance along an anthropomorphism continuum. The realistic avatar incorporated photorealistic visual design and high-fidelity motion animation, with specific attention to mitigating cues that intensify uncanny-valley responses, including subtle morphological inconsistencies and non-naturalistic movement. The stylized toon avatar, frequently used as a comparison to realistic human avatars in research on avatar effects (Combe et al., 2024; Fraser et al., 2024; Ring et al., 2014; Shin et al., 2019), conveyed clear anthropomorphic cues while deliberately reducing human realism. The robot avatar preserved a

humanoid structural outline but emphasized functional rather than biological anthropomorphism through simplified facial morphology and movement, resembling socially assistive robots used in mental health contexts (e.g., Woebot). Finally, the audio waveform animation served as a non-humanoid baseline, removing all social, facial, and bodily cues while still providing a visual stimulus comparable in modality to the embodied conditions. Prior work suggests that real-time, audio-driven visualizations can enhance task performance and perceptual processing relative to less informative or unsynchronized displays (Cartwright et al., 2017; Erdmann et al., 2025; Huang et al., 2025), supporting the use of waveform animations as a sufficiently engaging and robust control condition. The effectiveness of the intervention was assessed through self-report measures, including emotional ratings collected throughout the intervention and responses to an intervention satisfaction questionnaire administered at the end of the experiment. Additional key outcomes included participants' ratings of their impression of the avatar, their level of self-disclosure as well as open feedback on the digital intervention.

Drawing on the literature that highlights the positive effects of heightened anthropomorphism in avatars within learning and medical coaching contexts (Chae et al., 2024; Ring et al., 2014) and on the Perceptual Mismatch Hypothesis, which suggests that close alignment between appearance and behavior can mitigate uncanny-valley effects, it is hypothesized that avatars created with state-of-the-art photo-realistic digital-human technologies will yield superior results, leading to greater intervention effectiveness, a more positive avatar interaction experience, and higher levels of subjective self-disclosure.

2. Methods

2.1. Participant recruitment and sample characteristics

Participants were recruited via university-wide email invitations distributed by the University of Tübingen and the Technical University of Darmstadt as well as through online research platforms such as SurveyCircle and Psychologie Heute. The recruitment targeted adults interested in enhancing their emotion regulation and willing to participate in a study examining the effects of a brief, online-delivered digital emotion regulation intervention. As an incentive, participants were offered the opportunity to enter a lottery for a 25€ Amazon voucher or, for psychology students at the University of Tübingen, the chance to earn participant hours required for their coursework. Participants' privacy was protected, and all gave informed consent to take part. The study was conducted in compliance with ethical guidelines approved by the local ethics committee of the Institute for Psychology at the University of Tübingen (Kommission für Ethik in der psychologischen Forschung, approval date: 03/01/24, approval number: Az Naumann_2024_0130_282) and was pre-registered (<https://doi.org/10.17605/OSF.IO/CR2EY>).

A total of $n = 282$ individuals accessed the demographic questionnaire, which served as the initial step of the online experiment. Of these, $n = 233$ proceeded to the online intervention phase and $n = 212$ completed the entire study. One participant was excluded due to a response on a manipulation check item indicating insufficient engagement with the instructions, resulting in a final sample of $n = 211$. These participants were randomly assigned to the four experimental conditions: $n = 53$ to the realistic humanoid avatar condition (realistic HAC), $n = 54$ to the abstract toon-styled humanoid avatar condition (abstract HAC), $n = 52$ to the robot avatar condition (RAC), and $n = 52$ to the control avatar condition (CAC) with no embodied character but relying instead on an audio waveform animation to simulate interaction. Specific attrition rates following avatar introduction were as follows: $n = 5$ (9.4 %) in the realistic HAC condition, $n = 3$ (5.5 %) in the abstract HAC condition, $n = 6$ (11.5 %) in the RAC condition, and $n = 7$ (13.4 %) in the CAC condition.

2.2. Emotion regulation intervention

The present study utilized a brief, single-session emotion regulation intervention, lasting approximately 30–40 min. This training was administered digitally via video presentations, featuring a conversational avatar functioning as a digital coach. The intervention was designed to be interactive, employing a decision-tree algorithm that allowed the avatar's responses to dynamically adapt to participants' inputs. For instance, depending on the participant's self-reported current emotions, the avatar provided either positive reinforcement for a positive affective state or expressed empathy and encouragement in response to negative emotions.

The intervention comprised five structured phases: (a) an introductory segment, (b) psychoeducational content on the functions of emotions and adaptive emotion regulation strategies, (c) induction of negative emotion through a negative memory recall task to prepare participants for subsequent emotion regulation practice, (d) a practical exercise centered on cognitive reappraisal, and (e) a practical gratitude exercise, followed by closing remarks.

During the negative emotion induction phase, participants were encouraged to “open up” by reflecting on the negative emotional event they were asked to vividly recall, describing the situation, their emotions, thoughts, and perceptions in as much or as little detail as they wished in a free-text field. In the subsequent practical exercises, participants were similarly invited to record their reappraising, positive thoughts and adaptive strategies in a text field, again with full flexibility regarding the level of detail provided.

The present training was based on established emotion regulation manuals (Berking, 2017; Eismann & Lammers, 2017), which have been widely implemented in longer, group-based formats and shown to enhance emotional competence in both clinical and non-clinical populations (e.g., Berking & Lukas, 2015). In addition, the training incorporated core exercises targeting cognitive reappraisal and gratitude, strategies that research has shown to reduce stress and negative emotions while enhancing well-being in both experimental and applied settings (Diniz et al., 2023; Froh et al., 2008; Heckerens et al., 2022; Kalokerinos et al., 2015; Lee, Contreras Alejandro, & IJsselstein, 2024). An English translation of the exact wording used in the psychoeducational materials and adaptive emotion regulation exercises is provided in the Supplements.

The realistic and abstract human-like avatars were designed to be as visually similar as possible with regard to hair color, eye color, and clothing. The selection of avatar characteristics was informed by empirical findings suggesting that many clients prefer experienced, female therapists (Landes et al., 2013; Pikus & Heavey, 1996). All avatars as well as the control interface were created using specialized software platforms.¹ To ensure consistency across experimental conditions, voiceovers were generated using ElevenLabs, an AI-driven text-to-speech platform, using a female voice matched to the avatars' appearance and designed to convey emotional expressivity and a supportive interaction style, consistent with recommendations for effective virtual human interaction (Cihodaru-Ştefanache & Podina, 2025; Zhang et al., 2020).

¹ The realistic HAC stimuli were produced using HeyGen, the abstract HAC stimuli using Animaze, the RAC stimuli using Adobe Character Animator, and the CAC videos using VEED. All outputs were standardized with respect to script, duration, and audio parameters. Efforts were made to include emotionally expressive facial movements and non-verbal behaviors supporting socially engaging communication (e.g., tilting head, nods, smiles), in line with best practice recommendations for effective virtual humans (Cihodaru-Ştefanache & Podina, 2025; Zhang et al., 2020). To enhance perceived realism, the platforms allowed audio tracks to be integrated directly with automated lip syncing, ensuring that speech and mouth movements were closely aligned and comparable across conditions. Example videos for each condition are provided in the Supplementary Materials.

This approach was chosen to reflect real-world implementation of fully automated digital mental health interventions, where high-quality AI-generated speech enables scalable and consistent delivery across users and settings.

Additionally, subtitles were integrated into all presentations to enhance accessibility and clarity. Fig. 1a provides an overview of the avatars employed in each experimental condition.

2.3. Questionnaires and outcome measures

Questionnaires for Sample Description. (a) The Brief Symptom Inventory-18 (BSI-18; Derogatis, 2001; Franke, 2016) is a widely used self-report questionnaire designed to assess mental health symptoms. It consists of 18 items rated on a 5-point Likert scale (0 = “not at all” to 4 = “extremely”), where participants indicate the extent to which they have experienced specific symptoms over the past week. The BSI-18 provides subscale scores for depression, anxiety, and somatization, as well as a Global Severity Index (GSI), which represents overall psychological distress by summing all item scores. The BSI-18 has demonstrated excellent psychometric properties, with high internal consistency in the current sample (Cronbach's $\alpha = 0.93$). (b) The Difficulties in Emotion Regulation Scale – Short Form (DERS-SF; Kaufman et al., 2016) is a brief version of the original DERS (Gratz & Roemer, 2004), a widely used self-report measure of emotion regulation difficulties. The DERS-SF consists of 18 items rated on a 5-point Likert scale (1 = “almost never” to 5 = “almost always”) that assess six key dimensions of emotion dysregulation: nonacceptance of emotions, difficulties engaging in goal-directed behavior, impulse control difficulties, lack of emotional awareness, limited access to emotion regulation strategies, and lack of

emotional clarity. In addition to subscale scores, a total mean score can be calculated, with higher scores indicating greater difficulties in emotion regulation. The DERS-SF has strong psychometric properties and internal reliability in the current sample was high (Cronbach's $\alpha = 0.91$). (c) The Distress Disclosure Index (DDI; Kahn & Hessling, 2001) is a self-report questionnaire designed to measure an individual's tendency to disclose or conceal personal distress. It consists of 12 items rated on a 5-point Likert scale (1 = “strongly disagree” to 5 = “strongly agree”), assessing the extent to which individuals share their emotional struggles with others (e.g., “I am willing to tell others my distressing thoughts.”). Higher scores indicate a greater willingness to disclose distress, whereas lower scores suggest a tendency toward emotional concealment. The DDI is widely used in psychological research and demonstrated excellent internal consistency in the current sample (Cronbach's $\alpha = 0.94$).

Intervention Effectiveness. (a) Given that the primary objective of the intervention was to improve emotion regulation, repeated assessments of participants' momentary emotional states were incorporated as an index of intervention effectiveness and a means of tracking emotional change over the course of the training. Accordingly, participants were instructed to rate their current emotional state using a single-item 10-point Likert scale, ranging from 1 (“very negative”) to 10 (“very positive”). (b) The Client Satisfaction Questionnaire (CSQ-8; Attkisson & Zwick, 1982; Schmidt et al., 1989) is an 8-item self-report measure designed to assess client satisfaction with services received. Participants rate various aspects of service quality and overall satisfaction using a 4-point Likert scale (e.g., “How would you rate the quality of service you received?”; “In an overall, general sense, how satisfied are you with the service you received?”). Higher sum scores, potentially ranging from

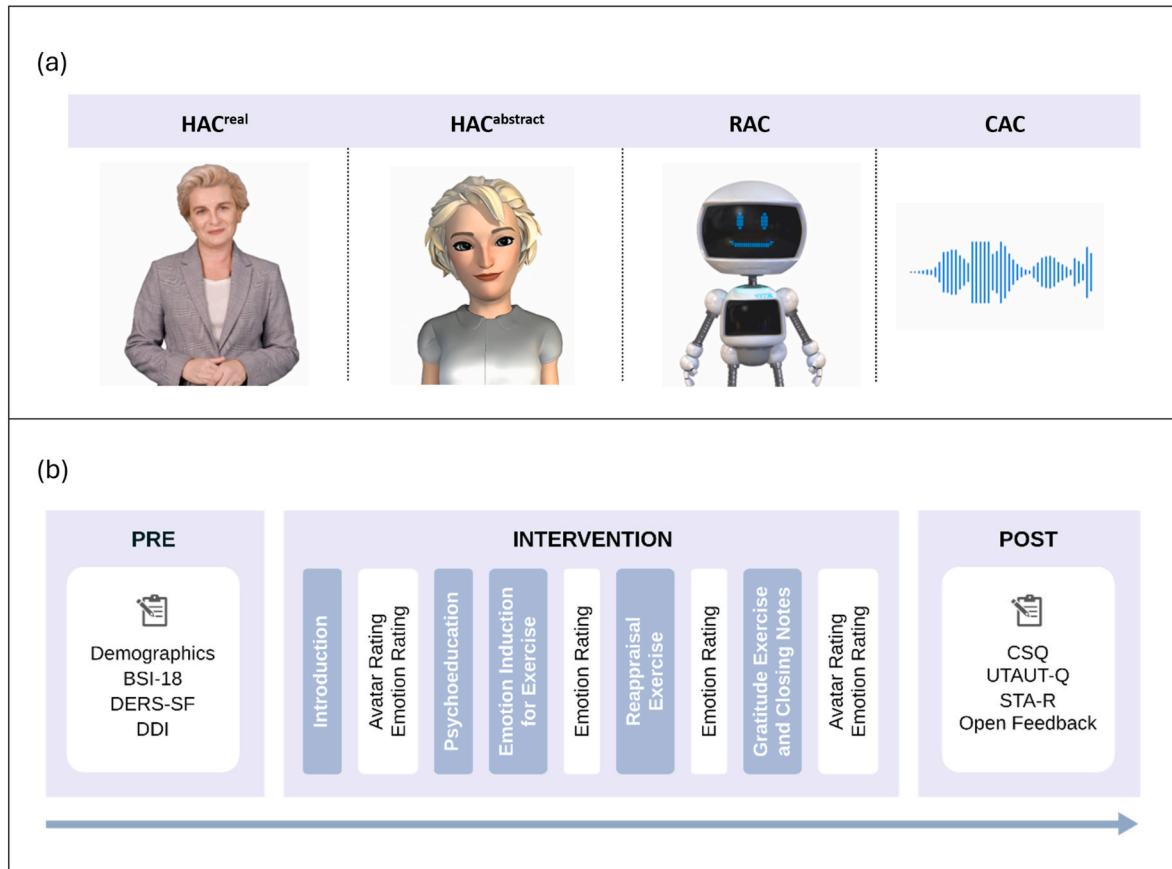


Fig. 1. Illustrations of (a) the experimental avatar conditions and (b) the overall experimental procedure.

Note. HAC^{real} = Realistic Humanoid Avatar Condition; HAC^{abstr} = Abstract Humanoid Avatar Condition; RAC = Robot Avatar Condition; CAC = Control Avatar Condition.

0 to 24, indicate greater satisfaction with the services provided. The CSQ-8 is widely utilized in evaluating the effectiveness of digital mental health interventions and clinical services more broadly. The instrument has demonstrated strong reliability and validity, with the current sample showing a high internal consistency (Cronbach's $\alpha = 0.92$).

Avatar Interaction Experience. (a) To evaluate participants' overall impression of the avatar and to explore potential changes in perception after extended interaction, participants were asked to rate their current overall impression of the avatar at the beginning and end of the training on a single 10-point Likert scale, ranging from 1 ("very negative") to 10 ("very positive"). (b) To assess perceived human-likeness of the avatar, participants responded to a single-item measure: "How human-like do you find the coach?" This item was included in the avatar impression ratings and rated on a 5-point Likert scale ranging from 0 ("not at all") to 4 ("very much"), administered at both the beginning and the end of the training. (c) Additionally, to assess participants' interaction experiences and impressions of the avatar, items and scales from the questionnaire by Heerink et al. (2010) were utilized. This questionnaire is based on the Unified Theory of Acceptance and Use of Technology (UTAUT, Venkatesh et al., 2003). Participants rated their interaction with the avatar on 16 items using 5-point Likert scales, ranging from 0 ("not at all") to 4 ("very much"), across multiple dimensions. These included a mean score on Anxiety (2 items; e.g., "I find the coach intimidating"; Cronbach's $\alpha = 0.84$), Pleasantness (4 items; e.g., "I think the coach is nice"; Cronbach's $\alpha = 0.86$), Enjoyment (5 items; e.g., "I enjoy the coach talking to me"; Cronbach's $\alpha = 0.87$), Usefulness (3 items; e.g., "I think the coach is useful to me"; Cronbach's $\alpha = 0.86$), and Trustworthiness (2 items; e.g., "I would follow the advice the coach gives me"; Cronbach's $\alpha = 0.77$).

Levels of Self-Disclosure. (a) A single self-developed rating item was used to assess participants' openness during the intervention. Participants rated the statement, "I was able to open up during the intervention," on a 5-point Likert scale ranging from 0 = "not at all" to 4 = "very much". (b) To evaluate participants' discomfort with disclosing personal information, the German STA-R questionnaire (Scale Therapeutic Alliance – Revised; Brockmann et al., 2011) was utilized. It is a validated measure of the therapeutic alliance based on widely established psychotherapy alliance questionnaires. For this study, two items from the Self-Disclosure Discomfort subscale were selected and adapted to fit the communication format of the digital intervention ("I felt too embarrassed to write about certain thoughts and feelings", "I found it difficult to openly reflect on my thoughts and feelings"). Both items were rated on a 5-point Likert scale (0 = "not at all" to 4 = "very much"). The correlation between these two items in the current sample, $r = 0.563$, $p < .001$, was used as an alternative measure of internal consistency, given the two-item structure.

Open Feedback. In line with participatory research efforts, participants were given the opportunity to provide open-ended feedback at the end of the session. They were encouraged to describe aspects of the intervention they found beneficial as well as areas they perceived as less effective.

2.4. Procedure

The study was conducted using the SoSci Survey software, which facilitated the presentation of questionnaires, stimulus materials, and avatar interaction videos associated with the emotion regulation intervention. Randomization into one of the four avatar conditions was performed using SoSci Survey's built-in randomization tool. Upon accessing the study, participants first reviewed the study information and provided informed consent. They then completed a series of trait questionnaires, namely the sociodemographic questionnaire (assessing gender, age, education, partnership status, and German language fluency), the BSI-18 (18 items), the DERS-SF (18 items), and the DDI (12 items), before proceeding to the intervention.

Before the initial intervention phase (i.e., after the brief introduction

by the avatar) and again at the end of the intervention, participants rated their current emotional state, overall impression of the avatar as well as its perceived human-likeness using the single-item measures. Additionally, emotion ratings were collected following the memory recall task involving a personal negative emotional experience (emotion induction) and after the subsequent reappraisal exercise.

After completing the intervention and the associated impression, human-likeness, and emotion ratings, participants filled out the CSQ (8 items), the UTAUT-based questionnaire (16 items), and self-disclosure ratings (single-item rating and the STA-R Disclosure Discomfort subscale with 2 items), and were given the opportunity to provide open-ended feedback. As a manipulation check in this fully anonymous online experiment, participants were asked to indicate whether they had genuinely read and followed the instructions. Finally, mental health support resources were provided to participants in case they required further assistance. A visual representation of the procedural steps can be found in Fig. 1b.

2.5. Statistical analyses

Data were analyzed using SPSS Statistics Version 28 (SPSS Inc., Chicago, IL). Two-way mixed-design ANOVAs were conducted to examine changes in emotion and avatar ratings over time. The between-subjects factor was Condition (four levels: realistic HAC, abstract HAC, RAC, CAC), and the within-subjects factor was Time, with four measurement points for emotion ratings (pre-intervention, post-emotion induction, post-reappraisal exercise, and post-intervention following the gratitude exercise) and two points for avatar ratings (pre- and post-intervention). When the assumption of sphericity was violated, as indicated by Mauchly's test, Greenhouse–Geisser corrections were applied to adjust degrees of freedom. Where a significant interaction between Condition and Time emerged, paired t -tests were performed within each condition to explore specific within-group changes.

For post-intervention-only outcomes (CSQ, UTAUT, and STA-R), one-way ANOVAs were used with Condition as the independent variable. In the event of significant main effects, post hoc comparisons were conducted using the Fisher's Least Significant Difference (LSD) test.

Open-ended feedback was evaluated by calculating the mean number of positive and negative statements provided by participants and comparing these between conditions. In addition, qualitative feedback was descriptively analyzed to identify common themes. Statements were categorized and themes generated by the author, with the resulting categorizations and themes reviewed by a colleague as an informal quality check; any uncertainties were resolved through discussion.

All statistical tests were two-tailed, with the threshold for significance set at $p < .05$. A priori power analyses indicated that the final sample size was sufficient to detect small-to-medium effect sizes with 80 % power at a 95 % confidence level ($\alpha = .05$, $\beta = 0.20$).²

3. Results

3.1. Sample description

Analysis of demographic variables revealed no significant differences between the experimental groups in terms of age, $p = .947$, gender, $p = .344$, education level, $p = .331$, relationship status, $p = .373$, mental well-being as measured by the BSI-18, $p = .564$, trait emotion regulation skills as measured by the DERS-SF, $p = .388$, or distress self-disclosure

² Please note that while the preregistration anticipated a slightly larger sample ($n = 280$), the recruitment period specified in the preregistration was strictly followed. Notably, the original power analysis in the preregistration targeted the detection of small effects (Cohen's $f = 0.10$), however, the final analysis focused on slightly larger effects (Cohen's $f = 0.12$), which remained within an acceptable range for the intended level of statistical power.

tendencies as measured by the DDI, $p = .933$. Table 1 presents the means and standard deviations of these demographic variables.

3.2. Intervention effectiveness

A 4 (Condition) $\times$ 4 (Time) repeated measures ANOVA on emotion ratings yielded a significant main effect of Time, $F_{(3,621)} = 65.91$, $p < .001$, $\eta_p^2 = 0.242$, as well as a significant interaction of Condition $\times$ Time, $F_{(9,621)} = 2.28$, $p = .024$, $\eta_p^2 = 0.032$, with no significant main effect of Condition, $p = .631$. Follow-up paired t-tests conducted separately for each condition revealed a significant decrease in positive emotion in response to the negative emotion induction across all conditions, $t_s \geq 3.76$, $p_s < 0.001$. Additionally, a significant increase in positive emotion was observed following the application of the adaptive emotion regulation strategy, cognitive reappraisal, $t_s \leq -2.66$, $p_s \leq 0.010$. However, while the abstract HAC, RAC, and CAC conditions showed a further significant increase in positive emotion in response to the gratitude exercise, $t_s \geq -3.05$, $p_s \leq 0.004$, no such effect was observed in the condition with the realistic human avatar, $p = .709$. This indicates that, unlike the other conditions, the gratitude exercise failed to further enhance positive emotions only in the realistic human avatar condition. Ms and SDs are displayed in Fig. 2a.

Furthermore, a one-way ANOVA on the CSQ sum score revealed a significant main effect of Condition, $F_{(3,207)} = 3.28$, $p = .022$, $\eta_p^2 = 0.045$. The follow-up LSD tests revealed that overall satisfaction ratings for the training were significantly lower in the condition featuring the realistic human avatar, $p_s \leq 0.032$, compared to all other experimental conditions. No significant differences were observed among the three remaining conditions, $p_s \geq 0.522$. Ms and SDs are given in Fig. 2b.

3.3. Avatar interaction experience

A 4 (Condition) $\times$ 2 (Time) repeated measures ANOVA on the overall impression rating of the avatar coach revealed a non-significant interaction effect, $p = .221$, but a significant main effect of Condition, $F_{(3,207)} = 6.47$, $p < .001$, $\eta_p^2 = 0.086$. Specifically, the realistic HAC received significantly more negative ratings compared to the other conditions, $p_s \leq 0.003$. Additionally, there was a significant main effect of Time, $F_{(1,207)} = 10.45$, $p = .001$, $\eta_p^2 = 0.048$, indicating an increase in positive

impressions over time. Ms and SDs are presented in Fig. 3a.

A repeated measures ANOVA on human-likeness ratings showed a significant interaction of Condition and Time, $F_{(3,207)} = 2.69$, $p = .047$, $\eta_p^2 = 0.038$, as well as a main effect of Condition, $F_{(3,207)} = 3.81$, $p = .011$, $\eta_p^2 = 0.052$, and a main effect of Time, $F_{(3,207)} = 12.79$, $p < .001$, $\eta_p^2 = 0.058$. Follow-up analyses comparing conditions at each time point revealed a trend for the realistic HAC and CAC to be perceived as more human-like than the abstract HAC and RAC at the beginning, $p_s \leq 0.056$. However, these differences were no longer observed at the end of the intervention, $p_s \geq 0.572$ (see Fig. 3b for Ms and SDs).

A one-way ANOVA on the UTAUT Anxiety score revealed a significant main effect of Condition, $F_{(3,207)} = 8.22$, $p < .001$, $\eta_p^2 = 0.107$. Follow-up LSD tests indicated that the realistic human avatar elicited significantly higher anxiety ratings, $p_s \leq 0.002$, compared to the other three conditions, which did not significantly differ from one another, $p_s \geq 0.105$. Similarly, the Condition effect on the UTAUT Pleasantness score was significant, $F_{(3,207)} = 2.66$, $p = .049$, $\eta_p^2 = 0.04$. LSD tests showed that interactions with the realistic human avatar were rated as significantly less pleasant than interactions with the other conditions, $p_s \leq 0.036$, while the other conditions did not significantly differ from each other, $p_s \geq 0.647$. However, ANOVA results for the UTAUT Enjoyment score, $p = .127$, the UTAUT Usefulness score, $p = .102$, and UTAUT Trustworthiness score, $p = .409$, did not indicate significant Condition effects. Ms and SDs are presented in Table 2.

3.4. Level of self-disclosure

A one-way ANOVA on the self-disclosure rating yielded a significant main effect of Condition, $F_{(3,207)} = 3.58$, $p = .015$, $\eta_p^2 = 0.049$. The follow-up LSD tests indicated that participants reported significantly lower levels of self-disclosure during the session in the condition with the realistic human avatar compared to all other conditions, $p_s \leq 0.012$. No significant differences were found among the three remaining conditions, $p_s \geq 0.866$.

There was no significant effect of Condition on the Disclosure Discomfort Scale, as measured by adapted items from the STA-R, $p = .362$. Ms and SDs are presented in Table 2.

3.5. Open feedback

A one-way ANOVA examining the average number of positive feedback points across intervention conditions did not yield a significant effect, $p = .707$. In contrast, the effect of Condition on the average number of negative feedback points was significant, $F_{(3,207)} = 4.57$, $p = .004$, $\eta_p^2 = 0.062$. Post hoc LSD tests revealed that the realistic avatar condition elicited significantly more negative feedback than all other conditions, $p_s \leq 0.009$, whereas no significant differences emerged among the remaining conditions, $p_s \geq 0.564$.

Participants provided a range of positive open-ended feedback on the emotion regulation intervention. The scalability and accessibility of the digital format were highlighted as strengths. Several participants noted that interacting with a digital coach, rather than a real person, fostered greater openness and emotional disclosure by reducing concerns about judgment or burdening others. The intervention's explicit clarification that it was not intended to replace traditional psychotherapy was also well-received, viewed as a responsible statement. Positive perceptions of the digital coach extended to its pleasant tone of voice. The delivery style was described as personable, empathetic, validating, and responsive. The content of the intervention was generally regarded as helpful, easy to understand, and delivered at a suitable pace. Exercises were perceived as feasible and well-structured, and participants particularly valued elements such as guided self-reflection and opportunities to apply the material to their personal experiences.

In addition to the positive feedback, participants also identified several areas for improvement. While many appreciated the avatar's tone of voice and speaking pace, others perceived these same features

Table 1
Means and standard deviations of demographic variables by condition.

		HAC ^{real} (n = 53)	HAC ^{abstr} (n = 54)	RAC (n = 52)	CAC (n = 52)
Age		28.49 (12.28)	29.22 (11.74)	27.92 (11.06)	28.31 (10.25)
Gender	♀	42	42	35	39
	♂	9	11	16	9
	* ○	2	1	1	4
Education	Medium	1	5	1	2
	High	52	49	51	50
Partnership	Yes	31	24	31	29
	No	22	30	21	23
BSI-18 - GSI		18.58 (13.44)	18.94 (15.23)	15.61 (10.89)	18.51 (14.14)
		2.53 (0.82)	2.51 (0.87)	2.34 (0.53)	2.34 (0.69)
DDI		40.30 (11.33)	39.24 (11.87)	38.86 (11.51)	39.38 (11.88)

Note. HAC^{real} = Realistic Humanoid Avatar Condition; HAC^{abstr} = Abstract Humanoid Avatar Condition; RAC = Robot Avatar Condition; CAC = Control Avatar Condition; BSI-18 - GSI = Global Severity Index of Brief Symptom Inventory; DERS-SF = mean score of Difficulties in Emotion Regulation Scale - Short Form; DDI = sum score of Distress Disclosure Index.

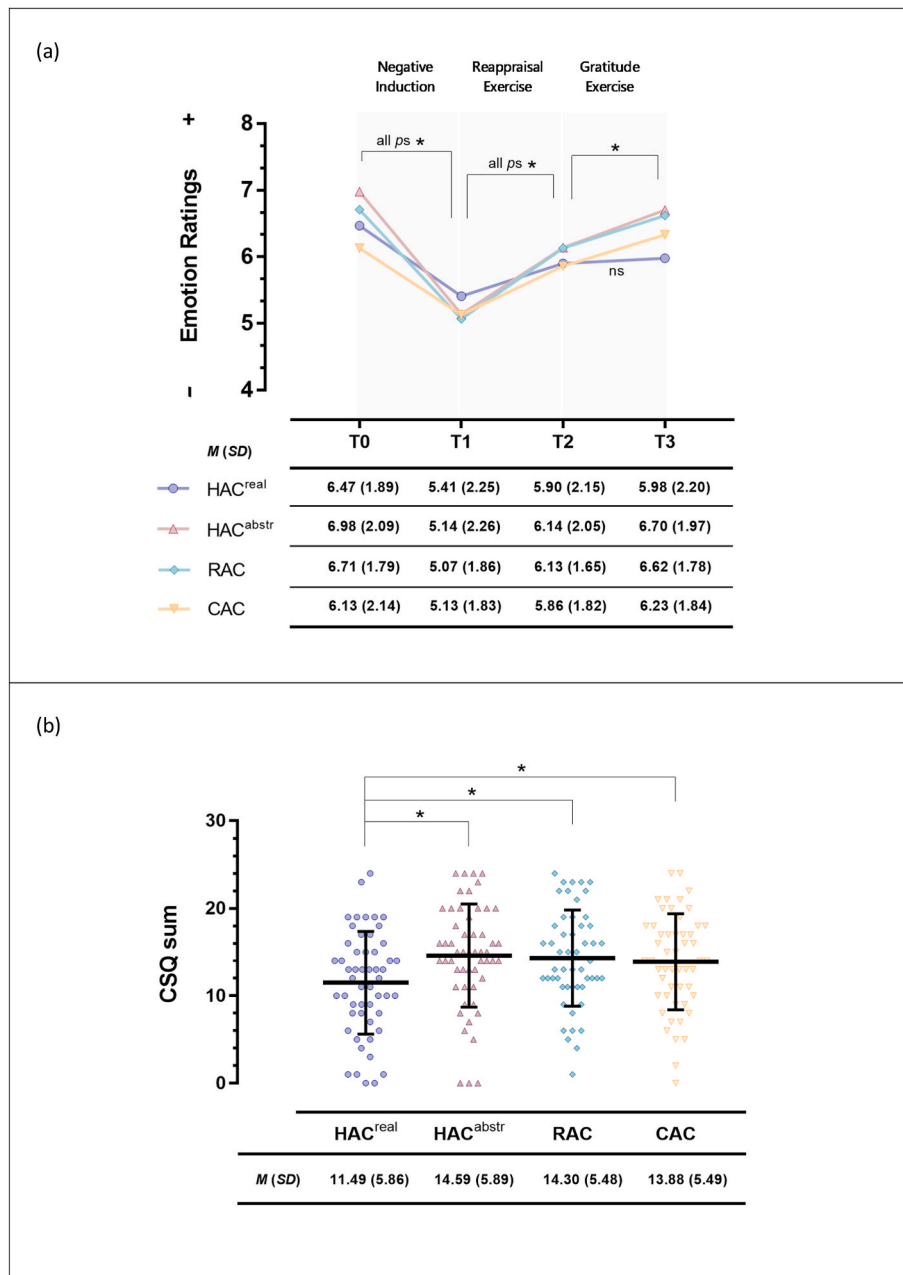


Fig. 2. Effectiveness of intervention: (a) emotion ratings and (b) client satisfaction (CSQ sum score).

less favorably. Specifically, some participants criticized the artificial voice, slow pace, and lack of synchrony between speech and lip movements, which detracted from the overall experience. Criticism related to appearance and presentation was particularly pronounced in the realistic humanoid avatar condition. Here, the avatar was often described as stiff, monotonous, or even eerie, occasionally with explicit references to the uncanny valley effect. In contrast, participants in the non-human avatar conditions sometimes expressed a preference for a more human-like appearance, indicating the need to carefully calibrate the degree of realism to enhance user comfort and engagement. Regarding the content, some participants felt that the material was overly general, too brief, or covered strategies with which they were already familiar. A very small number of participants found the exercises to be too challenging. Across conditions, a common theme in the feedback was a desire for more interactivity and a stronger sense of personalized responsiveness from the digital coach.

4. Discussion

This experimental study examined how avatar realism influences the effectiveness of a digital mental health intervention targeting emotion regulation. Participants were randomly assigned to one of four conditions, where the digital coach appeared as a realistic humanoid avatar, an abstract humanoid avatar, a robot avatar, or a sound-based animation. Intervention outcomes were assessed through repeated emotion ratings during the session, post-intervention satisfaction, evaluations of avatar interaction and self-disclosure, as well as open-ended feedback.

Across all conditions, participants successfully applied the reappraisal strategy during the first exercise, showing increased positive emotions without group differences. However, during the subsequent gratitude exercise, only those in the realistic humanoid condition failed to report further increases in positive emotion. It is important to note that while the training was based on established emotion regulation manuals and validated adaptive exercises, the single-session format used

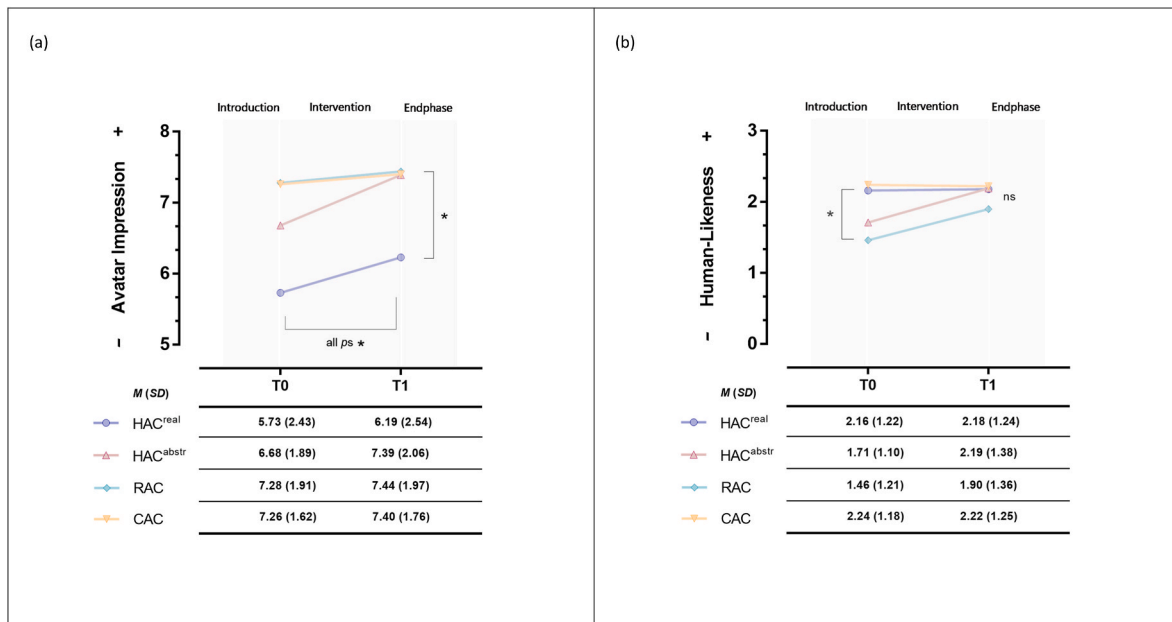


Fig. 3. (a) Overall impression ratings of avatar and (b) human-likeness ratings of avatar.

Table 2

Means and standard deviations for UTAUT questionnaire scales, self-disclosure levels, and open feedback count by condition.

	HAC ^{real} (n = 53)	HAC ^{abstr} (n = 54)	RAC (n = 52)	CAC (n = 52)
UTAUT Anxiety *	1.27 (1.18)	0.68 (0.94)	0.38 (0.71)	0.64 (0.88)
UTAUT Pleasantness *	2.43 (0.93)	2.80 (0.97)	2.88 (0.93)	2.81 (0.74)
UTAUT Enjoyment	2.24 (0.95)	2.47 (1.03)	2.54 (0.99)	2.68 (0.95)
UTAUT Usefulness	2.16 (1.05)	2.56 (0.96)	2.60 (1.05)	2.51 (0.94)
UTAUT	2.37 (1.02)	2.59 (0.93)	2.66 (0.92)	2.58 (0.73)
Trustworthiness				
Self-Disclosure Rating *	1.92 (1.22)	2.54 (1.21)	2.54 (1.19)	2.50 (0.98)
STA-R Disclosure	2.91 (2.57)	2.42 (1.91)	2.25 (1.65)	2.75 (2.11)
Discomfort				
Ø Number Positive Feedback	0.60 (0.96)	0.50 (0.81)	0.56 (0.80)	0.42 (0.69)
Ø Number Negative Feedback *	0.92 (1.15)	0.48 (0.77)	0.38 (0.77)	0.40 (0.66)

Note. Asterisks indicate significant effects of condition. HAC^{real} = Realistic Humanoid Avatar Condition; HAC^{abstr} = Abstract Humanoid Avatar Condition; RAC = Robot Avatar Condition; CAC = Control Avatar Condition; UTAUT = Unified Theory of Acceptance and Use of Technology, STA-R = Scale Therapeutic Alliance – Revised.

here has not been previously evaluated. In particular, evidence for the effectiveness of gratitude exercises largely comes from multi-day practice (Diniz et al., 2023), which may have constrained the impact of this component in the present study. Nevertheless, the fact that the other three conditions did not differ suggests that the observed effects can likely be attributed to the realistic humanoid avatar condition. This interpretation is further supported by participants' satisfaction ratings, where the realistic avatar received the lowest evaluations, while no differences were observed among the remaining conditions. These results suggest that human-like realism specifically reduced intervention effectiveness. These findings stand in contrast to the initial hypothesis and to prior evidence indicating that realistic human avatars outperform less human-like designs in education or medical contexts (e.g., Chae et al., 2024; Ring et al., 2014). One explanation may be that mental health interventions differ from purely instructional settings in requiring a strong collaborative bond, namely the therapeutic alliance, which has been shown to be a robust predictor of psychotherapy outcomes

(Flückiger et al., 2018). In such relationally demanding settings, highly realistic avatars may elevate expectations for natural social responsiveness; when these expectations are unmet, users may experience discomfort, aligning with both the Violation-of-Expectation and Perceptual Mismatch accounts of the uncanny valley (Cihodaru-Ştefanache & Podina, 2025; Zhang et al., 2020). In fact, in the present study the realistic avatar was initially rated as more human-like than the abstract and robot avatars, but did not differ from the audio waveform condition, and by the end of the session human-likeness ratings no longer differed across groups. Importantly, however, human-likeness was assessed using a single-item measure only and anthropomorphism is widely recognized as a multifaceted and partly subjective construct (Ho & MacDorman, 2017). Future research should therefore incorporate more comprehensive, multidimensional measures of anthropomorphism and assess user expectations early in the interaction, as recommended by Cihodaru-Ştefanache and Podina (2025), to more directly examine whether expectation-perception discrepancies contribute to the observed effects.

Qualitative feedback further highlights the nuanced role of human-likeness and expectation: several participants in the present study explicitly expressed a desire for more personalized and human-like interaction, highlighting the potential benefits of realism when paired with sufficiently responsive conversational capabilities. In the present intervention, however, the scripted, decision-tree interaction may have exacerbated perceived incongruence between the avatar's human-like appearance and its limited behavioral realism. This interpretation aligns with prior research showing that mismatches between avatar photorealism and behavioral realism can reduce social presence and interaction quality (Bailenson et al., 2005; Cihodaru-Ştefanache & Podina, 2025; Garau et al., 2003). Recent studies emphasize that, beyond photorealistic appearance in state-of-the-art digital humans, behavioral human-likeness must also be further developed to enhance user acceptance (Bae et al., 2025). Similarly, research on chatbots (non-embodied) has shown that contingent feedback elicits positive user responses, whereas generic feedback, particularly from an "expert"-labeled system, can produce negative effects, highlighting the importance of adequate emotional responsiveness and the detrimental impact of expectancy violations in social support contexts (Rheu et al., 2024). Future studies should investigate whether integrating adaptive, AI-driven dialogue systems into realistic avatars can reduce expectancy

violations and enhance intervention effectiveness in digital emotion regulation. Moreover, consistent with Yam et al. (2021), it may be valuable to explore whether pre-interaction instructions, such as framing the avatar in a more tool-like or intentionally dehumanized manner, can help align user expectations with the system's capabilities and mitigate potential negative responses.

Consistent with the uncanny valley hypothesis, the findings on the avatar interaction experience indicate that participants rated the realistic humanoid avatar least favorably overall, reported the highest levels of anxiety, and described it as the least pleasant to engage with. In contrast, the abstract humanoid, robot, and no-avatar control conditions did not differ significantly from one another. Open-ended feedback further underscored this pattern, with several participants explicitly referencing the uncanny valley effect or expressing feelings of eeriness toward the realistic humanoid avatar. As noted earlier, while some studies report advantages of realistic humanoid avatars, the present findings instead align more closely with research showing negative reactions to highly human-like designs. For example, Ciechanowski et al. (2019) found similar effects in chatbot interactions and Shin et al. (2019) showed that hyper-realistic avatars can even hinder information processing. Taken together, the broader literature suggests that avatar realism can have both positive and negative consequences depending on context, highlighting that design choices are not neutral but actively shape user experience and outcomes. Particularly for mental health interventions, it will be essential for future work to determine the conditions under which realism enhances versus undermines effectiveness.

Beyond visual appearance and behavioral realism, vocal characteristics constitute an additional and critical dimension of perceptual alignment in embodied agents. In the present study, all avatar conditions employed the same AI-generated voice to ensure experimental consistency and to reflect realistic deployment scenarios of fully automated digital mental health interventions. Although the selected voice was designed to convey emotional expressivity and a natural interaction style, and was positively evaluated as pleasant and supportive by many participants across conditions, a subset of users perceived the voice as artificial. Interpreted again through the lens of the perceptual mismatch account of the uncanny valley, such perceptions may be particularly consequential for avatars with high visual fidelity, as users are likely to hold elevated expectations regarding vocal naturalness and emotional prosody. When these expectations are not fully met, even subtle deviations may contribute to cross-modal incongruence and increased feelings of discomfort or eeriness. Accordingly, the present findings suggest that achieving sufficient alignment across visual, behavioral, and auditory channels may be necessary for highly realistic avatars to facilitate comfortable and effective user interactions in relationally sensitive contexts. These considerations align with recent findings emphasizing the interplay of facial expression and vocal delivery in shaping emotional perception and user experience (Suma et al., 2023). Future research may therefore benefit from systematically varying voice sources, including professional voice actors and differing levels of naturalness and emotional expressivity, to further clarify how vocal characteristics interact with visual realism and user expectations in shaping affective responses and avatar acceptance in digital mental health applications.

In addition to feelings of eeriness, participants' open feedback indicated that the realistic avatar was also perceived as stiff and boring. The choice of an elderly female avatar was based on earlier research suggesting preferences for experienced female therapists (Pikus & Heavey, 1996; Landes et al., 2013). However, this may have been a poor fit for this relatively young sample. Evidence suggests that younger users prefer avatars closer to their own age, especially in high-realism settings (Mikhailova et al., 2024). In general, individuals' preferences concerning a therapist's demographic profile are complex and multifaceted: some clients have been shown to dislike overly maternal figures (Russell et al., 2022) and gender preferences vary widely (Black & Gringart, 2019). Thus, it remains possible that a differently styled avatar would

have been evaluated more positively. Moreover, prior research indicates that attributes such as attractiveness and humor can mitigate uncanny valley responses, highlighting their relevance for future investigations (Cihodaru-Ştefanache & Podina, 2025; Rheu et al., 2021). Notably, when given the choice, the literature suggests that users prefer creating human-like over cartoon-like avatars and tend to bridge the uncanny valley by incorporating attractive, individualized features (Schwind et al., 2018). In this regard, emerging work further suggests that similarity between user and avatar can foster stronger bonds, greater self-disclosure, and higher satisfaction (Hooi & Cho, 2014; Pröbster et al., 2023). Yet, other findings show that uncanny valley effects may intensify when avatars resemble users in race (Saneyoshi et al., 2022) or facial features (Weisman & Peña, 2021). These results highlight that similarity alone does not ensure beneficial outcomes. This complexity parallels work on adolescent mental health app design, where participants expressed highly diverse and often incompatible preferences regarding interface aesthetics (Fleming et al., 2019; Kang & Kang, 2024). A promising strategy may, therefore, be to offer users a choice of avatars rather than assigning one by default. Supporting this, Xiao et al. (2007) found that letting participants choose their embodied conversational avatar improved impressions and persuasive impact, and Birk and Mandryk (2019) showed that avatar customization boosted engagement in attention bias trainings (see also Schwind et al., 2018). Furthermore, personalized avatars can better accommodate users' cultural and social backgrounds and preferences (Cihodaru-Ştefanache & Podina, 2025). Thus, future work should test whether allowing personalization in emotion regulation interventions with highly realistic avatars can strengthen social connectedness, reduce discomfort, and improve effectiveness. In this context, personalization may also need to encompass vocal characteristics, given individual differences in perceived vocal naturalness.

Consistent with the earlier findings, participants interacting with the realistic humanoid avatar reported the lowest self-disclosure, suggesting that increased anxiety and reduced pleasantness discouraged sharing. However, groups did not differ in reported discomfort with disclosure or in trust ratings toward the avatars, implying a motivational rather than affective barrier. Importantly, disclosure occurred in written format and may not reflect verbal disclosure in therapy. Again, the brief, single-session design further limits conclusions about longer-term dynamics. Prior research shows that repeated exposure to avatars and social robots can reduce eeriness, increase self-disclosure, and improve user experience (Laban et al., 2024; Schwind et al., 2018; Zlotowski et al., 2015). For instance, a recent study by Song and Shin (2024) found that increased avatar familiarity, analogous to the effect of celebrity recognition, attenuates the uncanny valley response to highly realistic avatars. In line with this, all avatars in the present study were rated more positively at the end of the session, reflecting a clear time effect. This suggests that initial negative impressions may fade with familiarity, making multi-session studies especially valuable. Regarding self-disclosure, participants in the present study frequently highlighted in open feedback that a key positive aspect of the digital emotion regulation session was the ability to share thoughts and feelings without fear of judgment or burdening another person. This underscores the potential of digital interventions as accessible, low-barrier mental health tools and aligns with prior findings on the advantages of digital trainings (Lee, Lee, & Lee, 2024; Papneja & Yadav, 2025). Supporting this, Lucas et al. (2014) provided experimental evidence that people are more willing to disclose to virtual humans believed to be fully automated than to those thought to be human-controlled. Evidence remains mixed on whether embodiment at all facilitates or hinders disclosure, with some studies favoring non-embodied designs (Papneja & Yadav, 2025). In the present study, no differences emerged between the abstract, robot, and sound-wave condition, though future work could include a no-embodiment control for clearer comparison.

Several limitations should be noted. First, as mentioned above, the intervention consisted of a single brief session with scripted, tree-based

interaction, limiting generalizability to longer or more interactive exchanges. Additionally, perceptions of human-like avatars are shaped not only by avatar characteristics but also by user-level factors such as novelty avoidance, neuroticism, or beliefs in the uniqueness of human nature (MacDorman & Entezari, 2015; Ratajczyk et al., 2024; Sasaki et al., 2017). The young, non-clinical sample further restricts applicability to older or clinical populations, whose responses to avatars may differ (Borghouts et al., 2021; Feng et al., 2018; Gillath et al., 2021; Tu et al., 2020). While the study was powered for medium effects, it may have been underpowered for smaller ones. Reliance on self-reports is another limitation, particularly for self-disclosure, which is often assessed more objectively through linguistic analyses and word counts in verbal disclosure contexts (Lucas et al., 2014). It would also have been informative to measure not only discomfort but also willingness to disclose to better differentiate between motivational and affective influences. As mentioned above, human-likeness was assessed with a single item, which limits sensitivity to its multifaceted and subjective nature. In particular, this measure did not allow for disentangling perceived human-likeness related to visual appearance, vocal characteristics, or behavioral responsiveness. Future studies should employ more comprehensive measures and assess user expectations early in the interaction to better capture potential effects of heightened anthropomorphism. In this context, relying on a unidimensional bipolar scale to assess emotional responses may have limited sensitivity to subtle changes in positive and negative affect. While research suggests that affective systems may shift toward stronger bipolarity in response to personally relevant or concerning events (Dejonckheere et al., 2021), there is also evidence that positive and negative affect can function as relatively independent constructs under certain conditions, exhibiting distinct dynamic patterns (Cooke et al., 2022). Accordingly, employing separate scales or a more comprehensive measure, such as the Positive and Negative Affect Schedule (PANAS; Watson et al., 1988), could allow for a more nuanced assessment of emotional changes and improve the precision of digital emotion regulation interventions. Moreover, individuals' responses to avatars are often not purely positive or negative, as people can perceive the same humanoid robot as both likable and frightening simultaneously (Kim et al., 2022). Consequently, future research should adopt more fine-grained measures to capture uncanny valley responses, assessing a wider spectrum of emotional reactions beyond anxiety, pleasantness, and enjoyment (Zhang et al., 2020). Furthermore, the therapeutic alliance with fully automated programs remains an underexplored construct. Existing measures are largely adapted from face-to-face therapy and may not fully capture the nuances of human-AI interaction (Tong et al., 2022; Tremain et al., 2020). Technical factors may have also contributed to participants' experiences. Although audio-visual synchronization was carefully managed, a few participants reported minor lip-sync or rendering delays, likely influenced by device or internet variability. While infrequent and distributed across conditions, such discrepancies could be especially salient in the realistic humanoid condition, where small expectation violations may exacerbate discomfort. These considerations highlight the value of controlled laboratory-based designs with standardized equipment. In addition, although many participants responded positively to the AI-generated voice in the present study, some noted overly synthetic qualities, underscoring the challenge of achieving high perceived realism. Future work should systematically disentangle the relative contributions of visual realism, vocal characteristics, and behavioral responsiveness, for example by experimentally manipulating audio-visual congruence and vocal naturalness within highly realistic avatars in digital mental health contexts. Finally, although the between-subjects design is a strength, a within-subjects design could potentially better reduce individual variability and, therefore, isolate avatar effects.

Despite these limitations, this study provides novel insights into avatar realism in digital mental health. The findings suggest that in relationally sensitive contexts such as emotion regulation, less realistic

or non-human-like avatars may be more effective, particularly when alignment across visual, vocal, and behavioral cues is suboptimal in highly realistic avatars. Future research should examine how personalization, repeated interaction, and improved multimodal realism can harness the potential appeal of human-like avatars while minimizing anxiety and discomfort, ultimately optimizing therapeutic outcomes.

5. Conclusion

This study investigated the role of avatar realism in a brief, fully automated digital emotion regulation intervention by experimentally comparing a highly realistic humanoid avatar with less realistic, more stylized embodied alternatives. Across conditions, participants were able to engage with the intervention and apply adaptive emotion regulation strategies; however, the highly realistic humanoid avatar was associated with less favorable user experiences, including higher anxiety, lower satisfaction, reduced self-disclosure, and no additional gains in positive affect following the gratitude exercise.

Taken together, the findings suggest that in relationally sensitive contexts such as emotion regulation interventions, heightened visual realism can sometimes backfire: by raising expectations for social and emotional responsiveness, highly realistic avatars may elicit discomfort and reduce perceived effectiveness when those expectations are not fully met, consistent with uncanny valley and expectancy mismatch accounts. These results should be interpreted in light of the study's scope as a brief, single-session, fully scripted intervention relying on self-report measures with a young, non-clinical sample, which may limit the generalizability of the findings to longer-term, more interactive, or clinical applications. From an implementation perspective, the findings emphasize the importance of carefully calibrating visual realism with vocal delivery and behavioral responsiveness to minimize expectancy violations and support positive user experiences. Future research can build on these insights by investigating how adaptive dialogue systems, richer behavioral realism, and multimodal synchronization influence user experience and therapeutic processes with human-like avatars across repeated sessions, incorporating both objective measures and clinical populations. Moreover, allowing users greater agency through avatar choice and personalization may help align expectations, reduce discomfort, and foster engagement and self-disclosure. Clarifying these factors will be essential for determining if, when, and how highly realistic avatars can be used most effectively in digital emotion regulation and broader mental health interventions.

Declaration of the use of AI

During the preparation of this work the author used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Declaration of competing interest

The author declares that she has no known competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chbr.2026.100952>.

Data availability

Data will be made available on request.

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